llmXive Automated Review of Scaling Mixture-of-Experts Video Pretraining for Embodied Intelligence

llmXive automated review panel

Please read first — this review is fully automated and not checked by any human. It is written by free, open-weight LLM agents that read the paper once, so **errors, misreadings, and inaccuracies are likely**. Nothing here has been verified by a person. Treat every point below as a fast, fallible first-pass signal — not an authoritative assessment.

How this review was produced

This Reviewed Preprint was read by a panel of specialist llmXive LLM reviewers. Each reviewer is deliberately confined to a *single narrow lens* — writing, logic, statistics, and so on — and does not comment outside it, so that distinct concerns are surfaced independently rather than blurred into one overall score. The panel runs *once* over the paper. Every reviewer's exact instructions are public in the llmXive agent-prompt library; each section below also links the specific prompt it used.

This report is **advisory, automated feedback for the authors**. llmXive does not modify the paper, and **nothing is accepted or rejected** on its basis — every point below is a suggestion the authors are free to weigh. Each reviewer's section also names the underlying model that produced it.

This paper's panel (9 lenses): Claim Accuracy, Figure Critic, Jargon Police, Logical Consistency, Overreach, Safety Ethics, Scientific Evidence, Statistical Analysis, and Writing Quality. Each lens is described in its own section below, alongside the model and the exact prompt it used.

Claim Accuracy

Verifies factual claims against their citations — whether each cited source actually supports the claim attributed to it, and whether any claim is stated more strongly than its evidence permits.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_claim_accuracy.md`

The paper makes several strong comparative claims in Section 6 (Evaluation) that require tighter alignment with the provided data. Specifically, the claim of “state-of-the-art” performance in Section 6.1 is slightly ambiguous regarding the inclusion of closed-source models (Wan~2.6, Seedance, Veo), which are listed in Table 1 but not explicitly excluded in the text’s summary of the TI2V results. While the authors likely mean “SOTA among open-source,” the phrasing “securing the top spot” could mislead a reader into thinking it beats all listed baselines.

Furthermore, the specific numerical claim of a “Physics-IQ Verified score of 40.4” in Section 6.2 is not present in any table or figure caption in the provided text; it appears only in the prose. Given that the paper relies heavily on visualizations (bar charts) for benchmark results, this specific number should be explicitly tabulated or cited from a specific figure data point to ensure reproducibility and verification. The comparison to Cosmos~3 (39.5) also needs to be clearly distinguished from the RBench scores in Table 1 to avoid confusion between different benchmark metrics.

Finally, the assertion in Section 5.1 that the FP8 “Speed-First” serving mode has “limited observed impact on video generation quality” is a load-bearing claim for the paper’s efficiency argument. However, Section 6 does not present a specific quality degradation metric (e.g., a drop in FID or a specific user study result comparing FP8 vs. BF16 outputs) to substantiate this. Without this data, the claim remains an unsupported qualitative observation. These issues are primarily matters of precision and evidence alignment rather than fundamental scientific flaws, warranting a minor revision to clarify the scope of claims and ensure all quantitative assertions are backed by visible data.

Action items.

- [**writing**] The paper makes several strong comparative claims in Section 6 (Evaluation) that require tighter alignment with the provided data. Specifically, the claim of “state-of-the-art” performance in Section 6.1 is slightly ambiguous regarding the inclusion of closed-source models (Wan~2.6, Seedance, Veo), which are listed in Table 1 but not explicitly excluded in the text’s summary of the TI2V results. While the authors likely mean “SOTA among open-source,” the phrasing “securing the top spot” could mi

Figure Critic

Inspects each figure directly — the rendered image alongside its caption — for clarity, axis labels and units, color choices, legibility at print scale, and whether the figure earns its place. This lens runs on a vision-capable model.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_figure_critic.md`

Figure 1

The figure serves as a visual teaser with high-quality samples, but the caption contains a grammatical error with a missing subject and fails to distinguish between the static images shown and the claimed video generation capabilities.

Figure 2

The figure provides a clear overview of the architecture, but the legend is incomplete as it omits definitions for the 'Attention' and 'Gate' blocks shown in the diagram. Additionally, the connection between the 'Time Embedding' and the modulation points is implied by proximity rather than explicit arrows.

Figure 3

The figure clearly displays training loss curves for three MoE configurations with appropriate legends and error shading, but the caption is overly generic and fails to describe the specific experimental comparison being visualized.

Figure 4

The figure presents a training loss comparison but suffers from a generic caption that fails to define the specific models or ablation parameters, and the truncated y-axis scale exaggerates the visual difference between the curves.

Figure 5

The figure effectively compares the two models, but the caption's claim that validation curves are 'unsmoothed' contradicts the visual evidence of perfectly smooth lines in the right panel. Additionally, the non-uniform x-axis tick spacing reduces readability.

Figure 6

Figure 6 presents training loss curves for dense and sparse models with a zoomed-in inset, but the x-axis misalignment between main and inset plots and the legend's placement within the plot area reduce clarity and hinder direct comparison.

Figure 7

The figure effectively demonstrates visual fidelity improvements between base and refiner generations for both human and robot subjects. However, the caption's reference to 'video generation' is slightly misleading as the figure only presents static frames, and it fails to explicitly highlight the successful text rendering of 'AURORA' as a key result.

Figure 8

Figure 8 effectively visualizes the Data Profiling Engine workflow described in the caption. The diagram clearly maps input media types to five annotated dimensions with specific examples, and the layout is uncluttered and easy to follow.

Figure 9

Figure 9 effectively visualizes the hierarchical structure of the World-Knowledge Topological Graph and the feedback loop for data curation. However, the text in the JSON annotation block is too small to read comfortably, and the distribution charts lack necessary axis labels and scales.

Figure 10

The Sankey diagram effectively visualizes the data curriculum flow, but lacks definitions for the resolution labels (192p, 480p, etc.) and the visual representation of the image stream's final retention is ambiguous relative to the stated percentages.

Figure 11

The figure provides a clear visual comparison of video quality before and after post-training, but the caption makes specific claims about text rendering and object deformation that are not supported by the visual evidence in the provided image.

Figure 12

Figure 12 effectively demonstrates qualitative improvements in embodied scenarios by comparing 'Original' and 'Post-training' results across four distinct tasks. The visual evidence clearly supports the caption's claims regarding the resolution of artifacts like structural distortion and non-physical penetration.

Action items.

- [**writing**] Figure 1: The caption contains a grammatical error and missing subject ('generated by . can produce...'), likely due to a missing model name placeholder.
- [**writing**] Figure 1: The caption claims to show 'Text-to-Video' samples, but the rendered image is a static collage of still images with no indication of motion or video frames.
- [**writing**] Figure 2: The legend in the top-right corner defines 'Routed Expert' and 'Shared Expert' but does not define the purple 'Attention' block or the pink 'Gate' block, which are distinct components in the diagram.
- [**writing**] Figure 2: The 'Time Embedding' label is placed at the bottom of the diagram, but the arrows originating from it are not explicitly drawn to the specific modulation points (the 'C' circles) in the main block diagram, making the connection implicit rather than explicit.
- [**writing**] Figure 3: The caption 'Training loss' is too generic to describe the specific

comparison of MoE expert scales (7B, 13B, 25B) shown in the legend; it should explicitly state the figure compares training loss across different MoE configurations.

- [**writing**] Figure 3: The x-axis label 'Training Steps' uses 'k' notation (10k, 20k) without explicitly defining 'k' as 1,000 in the caption or axis text, which is a minor clarity issue.
- [**writing**] Figure 4: The caption 'Training loss' is insufficient; it fails to define the two model variants (MoE 13B-A1.4B-E128 vs MoE 13B-A1.5B-E64) shown in the legend or explain the specific 'active capacity' ablation being compared.
- [**science**] Figure 4: The y-axis lacks a zero baseline and uses a truncated scale (0.15–0.17), which visually exaggerates the performance difference between the two models.
- [**science**] Figure 5: The caption states 'validation curves are unsmoothed', but the right panel shows perfectly smooth curves with no noise, which is inconsistent with raw logged losses. The left panel shows raw data in the background, but the right panel does not, contradicting the description.
- [**writing**] Figure 5: The x-axis tick labels (5k, 20k, 40k, 60k) are non-uniformly spaced relative to the axis ticks, making it difficult to accurately read the step count for specific points on the curve.
- [**science**] Figure 6: The inset 'Long-run zoom' shows curves extending to 80k steps, but the main plot's x-axis stops at 60k. This creates a disconnect where the inset displays data not present in the main view, and the main plot's x-axis scale (10k-60k) does not align with the inset's (20k-80k), making it difficult to correlate the two views.
- [**writing**] Figure 6: The legend is placed inside the plot area and partially obscures the curves, particularly the 'MoE 13B-A1.4B' and 'MoE 30B-A3B' lines. Moving the legend outside or using a more compact layout would improve readability.
- [**writing**] Figure 7: The caption describes the content as 'video generation' and 'refined video generation', but the rendered image displays static frames (likely keyframes) without any indication of motion or temporal progression, which is misleading for a video paper.
- [**writing**] Figure 7: The caption claims the robot's name is 'AURORA' and the prompt includes this detail, but the image shows the text 'AURORA' on the robot's chest; however, the caption does not explicitly point out that the text rendering is a specific success metric being demonstrated, making the connection implicit.
- [**writing**] Figure 9: The JSON object on the left is rendered at a very small font size, making the specific keys and values difficult to read and verify against the caption's description of 'structured tags'.
- [**writing**] Figure 9: The 'Long-tail distribution' panel contains bar charts with no axis labels, units, or numerical scales, rendering the data purely illustrative rather than informative.
- [**science**] Figure 10: The top 'Image' stream shows a retention rate of 27% at Stage 3, but the band width visually narrows to near zero by Stage 5, implying total elimination, yet the

caption states percentages are relative to the initial pool without clarifying if the stream is fully dropped or just reduced further.

- [**writing**] Figure 10: The '192p', '480p', and '1080p' labels at the top are not defined in the caption or legend; it is unclear if these refer to resolution thresholds, data quality tiers, or sampling criteria.
- [**science**] Figure 11: The caption claims to demonstrate improvements in 'blurred or incorrect text rendering,' but the provided image contains no text elements (e.g., signs, labels, or typography) to support this specific claim.
- [**science**] Figure 11: The caption claims to show 'structural object deformation' resolution, yet the 'Original' images (e.g., the excavator and motorcyclist) appear structurally sound and do not exhibit the deformation described.

Jargon Police

Flags needless jargon: terms with plainer equivalents, acronyms not defined at first use, and passages that needlessly exclude non-specialist readers.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_jargon_police.md`

The paper is generally well-structured and uses standard terminology for the diffusion and transformer communities (e.g., "DiT," "RoPE," "SwiGLU," "FlashAttention"). However, there are a few instances where acronyms or notation are introduced without sufficient definition for a competent reader from an adjacent field (e.g., a systems researcher or a computer vision specialist not deeply embedded in the specific MoE video generation subfield).

Specifically, the acronym "LLM" appears in the Introduction without being spelled out in the body text, relying on the abstract for the definition, which breaks the self-contained nature of the section. Additionally, the symbol in the GRPO training equations (Section 5) is used without a textual definition, forcing the reader to infer its meaning from the equation structure alone. The specific naming convention for model variants (e.g., "MoE 13B-A1.4B-E128") is also used without an explicit gloss of the format, which could be opaque to outsiders. Finally, while "CPS" is defined, its specific operational distinction from standard sampling methods could be clarified with a brief explanatory clause to ensure immediate understanding.

Addressing these minor gaps will ensure the paper is fully accessible to the target "adjacent-field PhD" audience without requiring them to cross-reference the abstract or guess at notation.

Action items.

- [**writing**] Section 1, Introduction: The acronym 'LLM' appears in 'large language models (LLM)' but is not expanded at first use in the body text (it is only defined in the abstract). Expand to 'large language models (LLMs)' at the first occurrence in Section 1 to ensure self-containment for adjacent-field readers.
- [**writing**] Section 2, Model Architecture: The symbol ' ' is introduced in the equation for

timestep-balanced gradient reweighting (Section 5, subsection On-Policy GRPO Training) without a definition in the surrounding text. It is unclear if this is a gain factor, a temperature, or a constant. Add a clause defining (e.g., 'where _k is the transition gain at step k') immediately after the equation.

- **[writing]** Section 2, Model Architecture: The term 'MoE 13B-A1.4B-E128' is used as a specific model identifier in Section 2.3 without a prior gloss explaining the naming convention (Total Params - Active Params - Expert Count). Define this convention once when first introduced to prevent confusion for readers unfamiliar with this specific shorthand.
- **[writing]** Section 4, Infrastructure: The acronym 'FSDP' is used in 'fully sharded data parallelism (FSDP)' but the expansion is slightly non-standard (usually 'Fully Sharded Data Parallel' without the 'ism'). While minor, ensure the expansion matches the standard literature or define it explicitly as 'Fully Sharded Data Parallelism' to avoid ambiguity for readers from adjacent distributed systems fields.
- **[writing]** Section 5, Training: The term 'CPS' (Coefficients-Preserving Sampling) is introduced in the context of GRPO training. While the full name is given, the acronym 'CPS' is not standard in the broader diffusion literature (unlike DDIM or SDE). Ensure the definition is prominent and perhaps add a brief parenthetical explaining its specific role (e.g., '...CPS (a DDIM-style transition that preserves noise coefficients)...') to distinguish it from other sampling methods.

Logical Consistency

Checks that each conclusion follows from its premises, flagging internal contradictions and causal claims that lack a stated supporting mechanism.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_logical_consistency.md`

The paper presents a coherent argument for a Mixture-of-Experts video foundation model tailored for embodied intelligence. The logical flow from the architectural design (Section 2) to data curation (Section 3), training strategy (Section 5), and evaluation (Section 6) is generally consistent. The premises regarding the need for capacity-compute decoupling and physical grounding are well-supported by the proposed methods and results.

However, three minor logical inconsistencies or ambiguities were identified that require clarification:

1. **Ambiguity in "Shared Expert Isolation" (Section 2.2):** The text states the design incorporates "shared expert isolation" from DeepSeekMoE. However, Equation 1 defines the output as a sum of shared experts (always-on) and routed experts. The term "isolation" is non-standard and could be misinterpreted as a structural separation rather than the intended meaning: shared experts are excluded from the router's top-K selection. This phrasing creates a slight disconnect between the textual claim and the mathematical definition.

2. **Imprecise Description of Stage 5 (Section 5.1 vs. Section 3.4):** Section 5.1 describes Stage 5 as being "trained on video data" for high-resolution refinement. In contrast, Section 3.4 explicitly defines this stage as using a "small high-quality video refinement set at 1080p." The vagueness in Section 5.1 could lead a reader to incorrectly assume the broad video data from earlier stages is used, rather than the specific high-resolution subset. Aligning the terminology would strengthen the logical consistency of the training curriculum description.
3. **Inconsistent Model Naming in Evaluation (Section 6.1 vs. Table 1):** The text in Section 6.1 refers to "Cosmos" when comparing scores, while Table 1 explicitly lists "Cosmos3 Super." This inconsistency in naming creates a minor logical gap where the reader must infer that "Cosmos" refers to the specific "Cosmos3 Super" baseline. Consistent naming throughout the evaluation section is necessary to ensure the comparison is unambiguous.

These issues are primarily matters of precision and do not undermine the core validity of the paper's central argument.

Action items.

- [**writing**] Section 2.2 claims 'shared expert isolation' but Eq. 1 shows shared experts are always-on, not isolated from routing. Clarify that 'isolation' means exclusion from the router's top-K selection, not structural separation.
- [**writing**] Section 5.1 describes Stage 5 as 'trained on video data' for refinement, while Section 3.4 specifies a 'high-quality 1080p subset'. Align Section 5.1 to specify the subset to avoid implying broad data usage.
- [**writing**] Section 6.1 refers to 'Cosmos' while Table 1 lists 'Cosmos3 Super'. Use consistent naming (e.g., 'Cosmos 3 Super') throughout the evaluation section to prevent baseline confusion.

Overreach

Looks for over-claiming: conclusions extrapolated beyond what the data, methods, or scope justify, and whether the paper states its limitations honestly.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_overreach.md`

The paper exhibits a pattern of overreach where the scope of its claims in the Abstract, Introduction, and Conclusion exceeds the specific boundaries of the evidence provided in the Evaluation section.

First, the claim of being the "inaugural" or "first" large-scale open-source MoE video foundation model for embodied intelligence (Abstract, Introduction) is not fully supported. While the model outperforms the specific baselines listed in Section 6 (Cosmos, Wan, LongCat, etc.), the paper does not provide a comprehensive survey or proof that no other MoE video models for embodied intelligence exist outside this specific comparison set. A "first" claim requires evidence of non-existence across the entire field, not just superiority over a curated list of competitors.

This should be qualified to “first among the evaluated open-source models” or supported by a broader literature review.

Second, the Conclusion makes strong functional claims about the model’s readiness for deployment, stating it serves as a “Policy Evaluator” for “safety-critical” tasks and an “Action Planner” for “real-time decision-making.” The evidence provided is strictly limited to offline generation benchmarks (RBench, Physics-IQ) and qualitative visual samples. There is no data from real-world robot deployments, no analysis of safety-critical failure modes in a physical loop, and no demonstration of real-time inference speeds within a control system. These assertions project the model’s capabilities far beyond the “simulation” and “generation” scope actually demonstrated. The language should be hedged to reflect potential (e.g., “demonstrates promise as a simulator for…”) rather than asserting established utility in safety-critical or real-time contexts.

Finally, the claim that the architecture is “highly practical” for “long-context video generation” (Section 2) relies on inference benchmarks up to 1M tokens. While impressive, this does not necessarily translate to the “long-context” requirements of embodied intelligence (e.g., multi-hour operational logs) or the specific latency constraints of real-time robot control. The paper omits a discussion of these deployment-specific limitations, presenting a synthetic benchmark result as a general solution for the target domain. Adding a limitations paragraph that explicitly distinguishes between token-count scalability and real-world embodied deployment constraints would align the rhetoric with the evidence.

Action items.

- [**writing**] Abstract/Intro claim ‘first’ MoE video model for embodied intelligence. Evidence only compares against a specific subset of baselines (Section 6). Qualify to ‘first among evaluated open-source models’ or provide broader survey to support the universal ‘first’ claim.
- [**writing**] Conclusion claims model serves as ‘safety-critical’ policy evaluator and ‘real-time’ action planner. Evidence is limited to offline benchmarks (RBench, Physics-IQ) and qualitative samples. No real-world deployment or safety analysis exists. Hedge claims to ‘potential for’ or ‘demonstrates promise as’.
- [**writing**] Section 2 claims architecture is ‘highly practical’ for long-context embodied generation based on 1M token benchmarks. This does not prove real-time control loop viability or multi-hour operational stability. Add a limitation distinguishing token-count scalability from real-world deployment constraints.

Safety Ethics

Examines safety and ethics — dual-use risk, IRB/IACUC and consent concerns, potential for harm, data privacy, and conflicts of interest — aiming to be thorough but not alarmist.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_safety_ethics.md`

The paper presents a large-scale video foundation model (LingBot-Video) for embodied intelligence, utilizing Mixture-of-Experts (MoE) architecture and a diverse dataset including internet videos and robot-oriented footage. From a safety and ethics perspective, the work is low-risk.

The primary data sources described are "internet videos" and "robot-oriented footage" (VLA, navigation, egocentric). The paper details a "Data Profiling Engine" that uses VLMs to filter for quality, aesthetics, and synthetic content, but it does not explicitly claim to use private, sensitive, or personally identifiable information (PII) from human subjects. The inclusion of "egocentric videos" and "human interaction" data is standard for this field; however, the paper does not describe collecting new human-subject data via surveys or interviews, nor does it release a raw dataset containing faces or PII. The released artifacts are model checkpoints and code, not the training corpus itself, mitigating the risk of PII leakage from the training set.

The methodology focuses on generating video simulations for robotics planning and policy evaluation. While generative models have dual-use potential (e.g., creating deepfakes or disinformation), the paper's specific contribution is a model optimized for physical realism and embodied tasks, not for high-fidelity human impersonation or deception. The paper does not provide operational details for cyber-attacks, biological synthesis, or surveillance systems. The discussion of "Physical Plausibility" and "Human-Motion Consistency" rewards aims to improve safety in simulation, not to enable harmful manipulation.

There are no indications of undisclosed conflicts of interest, license violations regarding the released code (which points to standard repositories), or unmitigated risks specific to the described architecture. The paper does not require a specific ethics statement beyond standard practice for pre-training on public web data, which is implicitly handled by the filtering pipeline described. No action items are necessary.

Scientific Evidence

Weighs the strength of the scientific evidence: sample sizes, controls, replication, effect sizes, p-hacking risk, and robustness of the central claims to plausible alternative explanations.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_scientific_evidence.md`

The paper presents a compelling architecture and training pipeline, but the evidentiary support for its central claims regarding performance superiority and physical grounding relies heavily on internal benchmarks and qualitative results that lack necessary statistical rigor.

First, the primary quantitative evidence in Section 6.1 (Internal Benchmark) and Table 1 (RBench) is presented as single-point estimates. For the internal benchmark, the paper does not disclose the number of prompts used, the number of generation seeds per prompt, or the variance (standard deviation) across seeds. In generative modeling, performance can fluctuate significantly based on random initialization and prompt selection. Without reporting mean $\pm$ standard deviation over multiple seeds (e.g., 3-5) and a clear description of the prompt distribution, the reported "state-of-the-art" margins could easily be artifacts of a lucky seed or a favorable subset of prompts. The same concern applies to the RBench and Physics-IQ results;

while these are public benchmarks, the paper does not explicitly confirm that all baselines were evaluated with identical inference hyperparameters (e.g., CFG scale, sampling steps, resolution). If the proposed model benefited from extensive hyperparameter tuning while baselines were run with defaults, the observed gains would reflect tuning effort rather than the intrinsic superiority of the MoE architecture.

Second, the claim that post-training with Reinforcement Learning (RL) significantly enhances physical plausibility (Section 5.2) is supported almost exclusively by qualitative figures (Figures 5 and 6) and a user study. While the user study provides some quantitative backing, it is limited to pairwise comparisons on a specific set of prompts and does not isolate the specific contribution of the RL objective against a non-RL baseline on a standardized metric. The absence of automated, quantitative metrics (e.g., success rates on a held-out physical reasoning test set or automated physics violation scores) for the RL vs. pre-trained comparison makes it difficult to rule out that the observed improvements are due to random variation or the specific selection of examples shown. To substantiate the claim that the RL phase is necessary for physical grounding, the authors should provide a quantitative evaluation on a held-out test set with multiple seeds, demonstrating a statistically significant improvement over the pre-trained model.

Finally, the scaling experiments in Section 2.3 (Scaling Experiments) rely on training and validation loss curves to argue for scalability. While these curves show consistent trends, they do not directly measure the downstream task performance (e.g., video generation quality or physical reasoning) at different scales. The claim that the model scales predictably to 120B parameters is based on early-stage training trajectories that are "not trained to full convergence." Without reporting downstream metrics for the larger models, the evidence for their practical utility remains indirect.

To strengthen the paper, the authors should: (1) report variance (mean $\pm$ SD) for all benchmark results across multiple seeds; (2) explicitly state and control for inference hyperparameters in all baseline comparisons; and (3) provide quantitative, automated metrics for the RL post-training improvements on a held-out test set.

Action items.

- **[writing]** The paper presents a compelling architecture and training pipeline, but the evidentiary support for its central claims regarding performance superiority and physical grounding relies heavily on internal benchmarks and qualitative results that lack necessary statistical rigor. First, the primary quantitative evidence in Section 6.1 (Internal Benchmark) and Table 1 (RBench) is presented as single-point estimates. For the internal benchmark, the paper does not disclose the number of prompts used, t

Statistical Analysis

Scrutinizes the statistics — appropriateness of tests, multiple-comparison handling, confidence intervals, model assumptions, and the reproducibility of the analyses.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_statistical_analysis.md`

The statistical reporting in the evaluation and scaling sections lacks necessary measures of uncertainty and significance testing, which limits the ability to distinguish genuine performance gains from random variance.

In **Section 6.1 (Internal Benchmark)** and **Table 1 (RBench)**, the authors report precise point estimates (e.g., 0.620, 0.578) for model performance across various tasks. However, there is no accompanying measure of uncertainty, such as standard deviation (SD), standard error (SE), or confidence intervals (CI). In video generation benchmarks, scores can vary due to the stochastic nature of the diffusion process and the specific subset of prompts used. Without reporting variance (e.g., "0.620 $\pm$ 0.015 over 3 seeds"), readers cannot determine if the reported "state-of-the-art" margins are statistically robust or within the noise floor of the evaluation. This is a reporting gap that should be fixed by aggregating results from multiple evaluation runs.

In **Section 6.3 (User Study)**, the paper presents Good-Same-Bad (GSB) results based on 400 prompts per comparison pair. While the raw counts or percentages are implied, the text makes qualitative claims of superiority ("clearly outperforms," "consistent advantage") without performing or reporting a formal statistical test. For pairwise human evaluations, a binomial test or McNemar's test is standard to determine if the proportion of "Good" votes significantly exceeds the "Bad" votes (or 50% chance). Additionally, effect sizes (e.g., odds ratios) should be reported to quantify the magnitude of the improvement, rather than just stating it is "clear."

Finally, in **Section 5.2 (Scaling Experiments)**, the training loss curves (Figures 3, 4, 5) and inference latency ratios are presented as single, smooth lines. In deep learning research, it is standard practice to report these metrics with error bands (e.g., ± 1 SD) across multiple random seeds (typically 3-5) to demonstrate that the observed scaling laws and efficiency gains are reproducible and not artifacts of a specific random initialization or data shuffle. The absence of these error bands makes the "predictable scaling" claim harder to verify statistically.

These issues are primarily reporting gaps (severity: writing) that can be resolved by re-analyzing existing evaluation data or re-running the evaluation scripts with multiple seeds, rather than requiring new model training.

Action items.

- [**writing**] Table 1 (RBench) and Section 6.1 report point estimates (e.g., 0.620) for model performance without any measure of uncertainty (standard deviation, confidence interval, or range). Given that these scores likely derive from a finite set of prompts (650 for RBench) and potentially stochastic generation, the absence of variance reporting prevents assessment of result stability. Report mean $\pm$ SD over multiple evaluation runs or provide 95% confidence intervals for all benchmark scores.
- [**writing**] Section 6.3 (User Study) reports Good/Same/Bad rates based on 400 prompts per pair but omits statistical significance testing. Claims like 'clearly outperforms' or 'consistent advantage' are made without p-values or effect sizes (e.g., Cohen's d or odds ratios) derived from the pairwise comparisons. Apply a binomial test or McNemar's test to the GSB counts to determine if the observed 'Good' rates are statistically significant ($p < 0.05$) and report the effect size.
- [**writing**] Section 5.2.1 (Scaling Experiments) and Figures 3-5 present training/validation loss

curves and inference latency ratios as single deterministic lines. For deep learning experiments involving stochastic optimization and data sampling, these metrics should be reported with uncertainty bands (e.g., ± 1 standard deviation over 3-5 seeds) to distinguish genuine scaling trends from random variance. Add error bands to the loss curves and report variance for the latency ratios.

Writing Quality

Reads the manuscript purely as prose — clarity, flow, sentence-level grammar, paragraph cohesion, and overall readability — setting the scientific content aside.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_writing_quality.md`

The paper is generally well-structured, but several instances of grammatical errors, wordy phrasing, and weak topic sentences impede smooth reading.

In **Section 2.1**, the opening paragraph lists specific components before clearly stating the section's organizational structure. The sentence "We use Qwen3-VL-4B... to extract condition" is grammatically incomplete, missing the article "the" or the plural "conditions." A stronger topic sentence should introduce the architecture's purpose before listing its parts.

Section 2.2 contains a long, convoluted sentence in the first paragraph regarding video generation demands. This sentence is overloaded with clauses and should be split or simplified. Additionally, the paragraph mixes the motivation for MoE with a definition of dense models, which could be separated for better logical progression.

In **Section 2.3**, the topic sentence is buried. The paragraph begins with a long introductory clause ("To identify...") rather than the main action. A more direct opening would improve clarity.

Section 3.1 has a minor but noticeable flow issue where a figure reference is awkwardly placed mid-sentence with a tilde, breaking the reading rhythm. Integrating the reference more smoothly would be better.

Section 5.1 features a long list of risks that makes the sentence feel heavy. Breaking this into two sentences would improve readability.

Finally, **Section 6.1** contains a subject-verb agreement error: "the internal benchmark cover" should be "covers." The introductory phrase is also redundant given the context.

Addressing these specific grammatical and structural issues will significantly enhance the paper's readability.

Action items.

- [**writing**] Section 2.1, para 1: 'extract condition' is grammatically incomplete. Add 'the' or 'conditions'. Also, the paragraph lists components before stating the section's purpose. Rewrite to state the architecture's goal first, then list components.
- [**writing**] Section 2.2, para 1: The sentence 'Video generation, which aims to simulate...' is overloaded. Split it. Also, the paragraph mixes MoE motivation with dense model definitions;

separate these points for better flow.

- **[writing]** Section 2.3, para 1: The topic sentence is buried. Start with ‘We systematically explore the MoE design space...’ instead of the long ‘To identify...’ clause to improve clarity.
- **[writing]** Section 3.1, para 1: The figure reference ‘~\cref{fig:profiling_engine}’ breaks the sentence flow. Integrate it as ‘(see Fig. X)’ or move it to the end of the paragraph.
- **[writing]** Section 5.1, para 1: The list of risks (‘optimization instability, routing collapse...’) makes the sentence heavy. Split into two sentences for better readability.
- **[writing]** Section 6.1, para 1: ‘benchmark cover’ is a subject-verb agreement error; change to ‘covers’. Remove the redundant phrase ‘To comprehensively evaluate...’.